University of Engineering and Technology

Department of Mechatronics and Control Engineering

TERM PROJECT REPORT

Application of Information and Communication Technologies

Course Code: MCT-107L

Customer Complaint Tracking System

Group Members:

2025-MC-155

2025-MC-159

Instructor:

Engr. Syed Muhammad Umer

Semester: 1^{st} Semester

Session: 2025

Term Project – Complex Engineering Activity (CEA)

Course: Application of Information and Communication Technologies (AICT)
Course Code: MCT-107L
Instructor: Engr. Syed Muhammad Umer
Semester: 1st | Session: 2025

Project Overview

Each group will design and document an ICT-based information system addressing a real-world problem (e.g., attendance management, patient records, smart parking, inventory control, online booking). Students will create and analyse datasets, simulate system workflows, and justify design decisions using the ICT tools covered in the laboratory. Each project is unique and qualifies as a Complex Engineering Activity (CEA).

Required Tools

- Overleaf – LaTeX-based final report
- GitHub – Version control and project repository
- DBMS concepts – Database schema and sample SQL queries
- Google Sheets – Dataset creation, analysis, and simulation
- Canva – System diagram and project presentation
- Mendeley – Reference management and citations

Registration-Number-Based Project Allocation

The first group member’s registration number determines the project configuration.
Format: 2025-MCT-ABC (A, B, C are the last three digits).

Rule 1 – Project Domain (Digit B)

- 0 – Library Management System
- 1 – Hospital / Patient Record System
- 2 – School Attendance / LMS
- 3 – Transportation / Vehicle Log
- 4 – Inventory / Warehouse Management
- 5 – Customer Complaint Tracking
- 6 – Event Management & Registration
- 7 – Smart Parking / Campus Access
- 8 – Hostel / Accommodation Management
- 9 – Online Booking / Appointment System

Rule 2 – Dataset Size (Digit A)

Dataset size in Google Sheets = A × 10 records (e.g., A = 3 → 30 records, A = 8 → 80 records).

Rule 3 – Required Analysis (Digit C)

- 0 – Pivot table + Pie chart
- 1 – Line chart + Conditional formatting
- 2 – Forecast simulation (trendline)
- 3 – Dashboard creation
- 4 – VLOOKUP + MATCH
- 5 – DBMS schema + SQL queries
- 6 – Time-series simulation
- 7 – Comparison charts + Advanced filtering
- 8 – Multi-sheet system with cross-sheet formulas
- 9 – Mini Management Information System (MIS)

Project Tasks

- Problem identification and justification
- ICT system design and architecture
- Dataset creation and analysis in Google Sheets
- Basic DBMS schema design
- Documentation in Overleaf with Mendeley citations
- Version control and submission via GitHub

Deliverables

- GitHub repository link
- Google Sheets dataset and analysis
- Canva system diagram or infographic
- Final LaTeX report (Overleaf PDF)
- Group viva and presentation

Evaluation Criteria

Projects will be evaluated based on problem understanding, system design, quality of analysis, effective use of ICT tools, documentation quality, and viva/presentation performance.

Final Report Structure (Overleaf)

1. Title Page
2. Introduction
3. Problem Definition
4. Literature Review
5. Proposed System
6. Dataset and Analysis
7. DBMS Schema
8. System Diagram (Canva)
9. Results
10. Conclusion
11. References

Evaluation Rubric – Term Project (CEA) **Total Marks: 40**

Component	Excellent (9–10)	Good (7–8)	Satisfactory (5–6)	Poor (0–4)	Obtained
Dataset & Analysis (Google Sheets)	Dataset size fully correct; accurate formulas; required analysis completed; clear charts/dashboards; results well-interpreted.	Dataset mostly correct; analysis implemented with minor errors; charts present with limited interpretation.	Dataset present but incomplete or partially incorrect; basic analysis and charts only.	Dataset incorrect, missing, or no meaningful analysis performed.	
Use of ICT Tools	All required tools (Overleaf, GitHub, Canva, Mendeley) used correctly and effectively with clear integration.	All tools used with minor issues (e.g., limited GitHub commits, basic diagrams).	Some tools used properly; others used superficially or incorrectly.	One or more required tools missing or used incorrectly.	
Report Quality (Overleaf)	Well-structured LaTeX report; clear writing; logical flow; correct citations and references.	Complete report with minor formatting or clarity issues; references mostly correct.	Weak structure; unclear sections; formatting or citation issues present.	Poorly written or incomplete report; missing references or improper formatting.	
Viva & Presentation	All members demonstrate strong understanding; confident explanations and responses.	Most members show understanding; explanations adequate with minor gaps.	Limited understanding; uneven participation among members.	Weak understanding; inability to explain project or lack of participation.	
Total					

Important Grading Notes

All group members must participate in the viva.

The project must qualify as a **Complex Engineering Activity (CEA)**.

Plagiarism will result in **zero marks**.

Missing any required ICT tool will lead to **marks deduction**.

Late submissions will not be accepted.

Customer Complaint Tracking System

Introduction

Customer complaints are common in service organizations. Handling complaints manually is slow and causes confusion. Records can be lost and complaint status is not clear.

Information and Communication Technology (ICT) helps in managing complaints in an organized and efficient way. This project focuses on designing a Customer Complaint Tracking System using ICT tools.

Problem Definition

Many organizations manage customer complaints using paper records or unstructured files. This causes several problems such as delays, missing data, and poor tracking of complaint status.

There is a need for a computerized system that can store complaints properly, track their progress, and help staff resolve issues quickly.

Objectives of the Project

The objectives of this project are:

- To design a simple customer complaint tracking system
- To store complaint data in a structured database
- To track complaint status easily
- To use ICT tools for system development and documentation

Proposed System

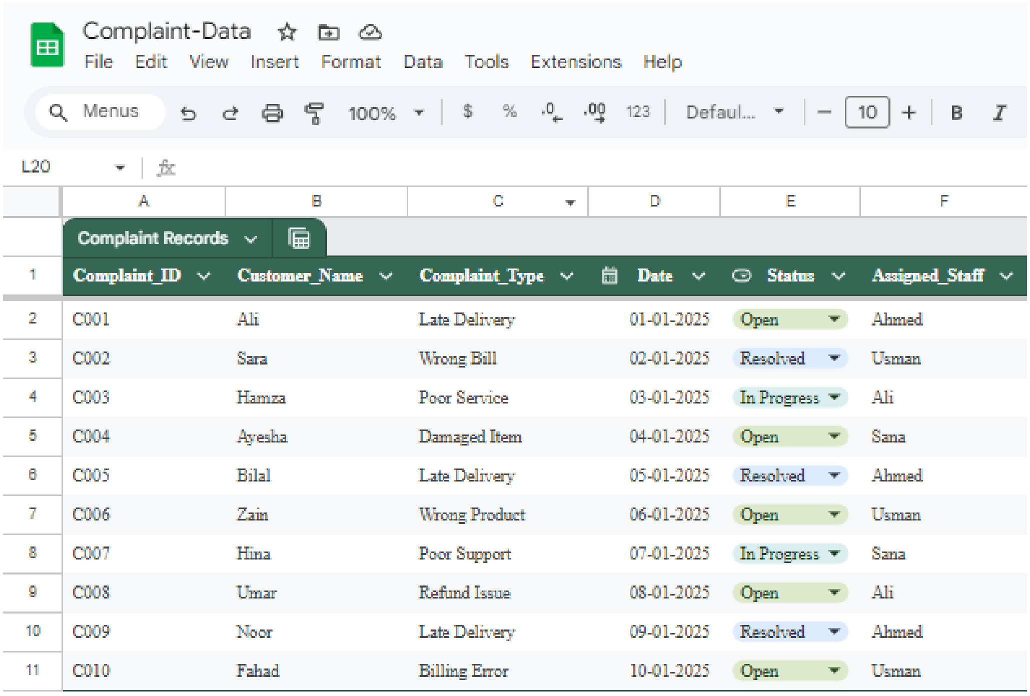
The proposed system allows customers to register complaints. These complaints are stored in a database and assigned to support staff. Each complaint has a status such as

Open, In-Progress, or Closed.

The system improves complaint handling, reduces delays, and keeps records organized.

Dataset Creation and Analysis

A dataset of 10 complaint records was created using Google Sheets according to the registration number rule. The dataset includes complaint ID, customer name, complaint type, date, status, and assigned staff.



Complaint_ID	Customer_Name	Complaint_Type	Date	Status	Assigned_Staff
C001	Ali	Late Delivery	01-01-2025	Open	Ahmed
C002	Sara	Wrong Bill	02-01-2025	Resolved	Usman
C003	Hamza	Poor Service	03-01-2025	In Progress	Ali
C004	Ayesha	Damaged Item	04-01-2025	Open	Sana
C005	Bilal	Late Delivery	05-01-2025	Resolved	Ahmed
C006	Zain	Wrong Product	06-01-2025	Open	Usman
C007	Hina	Poor Support	07-01-2025	In Progress	Sana
C008	Umar	Refund Issue	08-01-2025	Open	Ali
C009	Noor	Late Delivery	09-01-2025	Resolved	Ahmed
C010	Fahad	Billing Error	10-01-2025	Open	Usman

Figure 1.1: Complaint Data in Google Sheet

Database Design (DBMS Schema)

The database design includes three main tables:

- Customer Table – stores customer information
- Complaint Table – stores complaint details
- Support Staff Table – stores staff information

Primary keys and foreign keys are used to maintain relationships between tables.

complaint_id	customer_name	complaint_type	complaint_date	status	assigned_staff
1	Ali	Late Delivery	2025-01-01	Open	Ahmed
2	Sara	Wrong Bill	2025-01-02	Resolved	Usman
3	Hamza	Poor Service	2025-01-03	In Progress	Ali
4	Ayesha	Damaged Item	2025-01-04	Open	Sana
5	Bilal	Late Delivery	2025-01-05	Resolved	Ahmed
6	Zain	Wrong Product	2025-01-06	Open	Usman
7	Hina	Poor Support	2025-01-07	In Progress	Sana
8	Umar	Refund Issue	2025-01-08	Open	Ali
9	Noor	Late Delivery	2025-01-09	Resolved	Ahmed
10	Fahad	Billing Error	2025-01-10	Open	Usman

Figure 1.2: SQL Table

System Diagram

A system diagram was created using Canva. The diagram shows the flow of data from customer complaint submission to database storage and assignment to support staff.

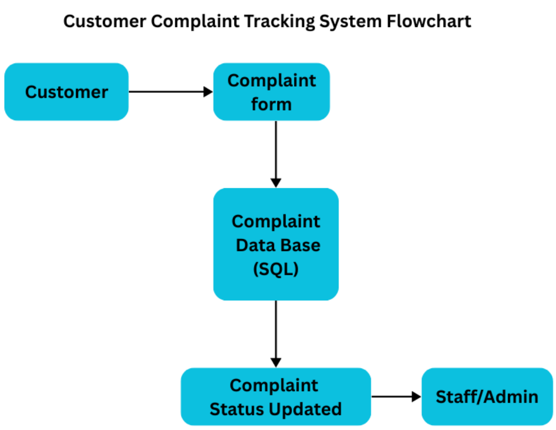


Figure 1.3: Complaint System Flowchart

ICT Tools Used

The following ICT tools were used in this project:

- Google Sheets for dataset creation
- DBMS concepts for database design

- Canva for system diagram
- Overleaf for report writing
- GitHub for project submission
- Mendeley for references

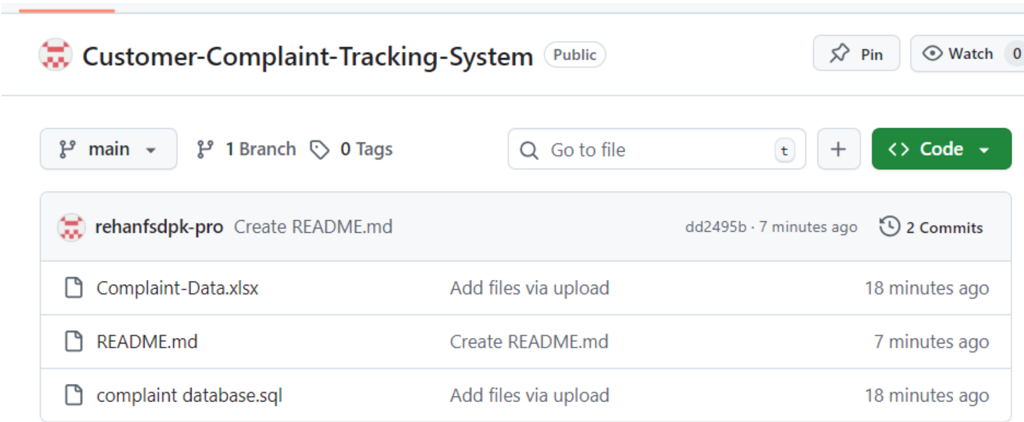


Figure 1.4: Github Repository

Results

The system provides organized complaint management, faster response, and better tracking of complaint status. It improves efficiency and reduces manual work.

Conclusion

The Customer Complaint Tracking System is an effective ICT-based solution for managing customer complaints [1]. The project fulfills the requirements of a Complex Engineering Activity (CEA) and demonstrates practical use of ICT tools in solving real-world problems.

References

[1] Laudon, K. C., & Laudon, J. P. Management Information Systems.